

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 17

Level 1 — Beginner · Geometry II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 1 — Beginner	Topic Focus	Geometry II — Perimeter & Area
Questions	20	Time Allowed	30 minutes
Total Marks	20	Worksheet No.	17 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 17 teaches the two most-confused ideas in primary geometry — perimeter (around) and area (inside) — and trains the child to know WHICH one a problem needs (fencing vs covering). Formulas for rectangles and squares, counting areas on grids, and reverse questions (perimeter given, find the side) build real Olympiad readiness. Four diagrams included.

Learning Objectives

- Distinguish perimeter (boundary) from area (inside space) and choose correctly.
- Use perimeter formulas for rectangles, squares and triangles.
- Use area = length × breadth and side × side, and count area on grids.
- Solve reverse problems: find a side from the perimeter or area.
- Apply perimeter and area to fences, walks, tiles and wires.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

**SECTION A · Multiple Choice Questions · 20 Questions**

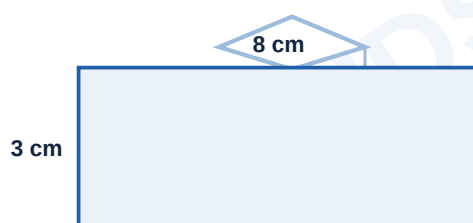
Q1. Perimeter of a shape means:

- (A) The space inside the shape (B) The total length of its boundary
(C) The number of corners (D) The height of the shape

Q2. A triangle has sides 5 cm, 6 cm and 7 cm. Its perimeter is:

- (A) 16 cm (B) 17 cm
(C) 18 cm (D) 19 cm

Q3. Find the perimeter of this rectangle.

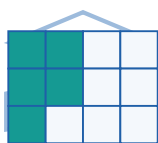


- (A) 11 cm (B) 22 cm
(C) 24 cm (D) 16 cm

Q4. A square has a side of 6 cm. Its perimeter is:

- (A) 12 cm (B) 18 cm
(C) 24 cm (D) 36 cm

Q5. Each small square in the grid is 1 square unit. What is the AREA of the shaded part?



- (A) 4 sq units (B) 5 sq units
(C) 6 sq units (D) 7 sq units

Q6. Area of a shape means:

- (A) The length of its boundary (B) The space covered INSIDE the shape
(C) The number of sides (D) The longest side

Q7. A rectangle is 7 cm long and 4 cm wide. Its AREA is:

- (A) 11 sq cm (B) 22 sq cm
(C) 28 sq cm (D) 24 sq cm



Q8. A square has a side of 5 cm. Its AREA is:

- (A) 10 sq cm (B) 20 sq cm
(C) 25 sq cm (D) 30 sq cm

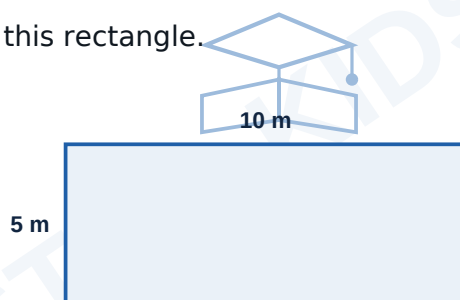
Q9. A farmer wants to put a FENCE around his rectangular field. Which should he calculate?

- (A) Area (B) Perimeter
(C) Volume (D) Height

Q10. Mina wants to cover her tabletop completely with paper. Which should she calculate?

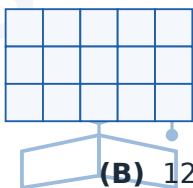
- (A) Perimeter (B) Area
(C) Number of corners (D) Length only

Q11. Find the perimeter of this rectangle.



- (A) 15 m (B) 25 m
(C) 30 m (D) 50 m

Q12. Each small square is 1 square unit. What is the area of the FULL grid (all squares)?



- (A) 8 sq units (B) 12 sq units
(C) 15 sq units (D) 16 sq units

Q13. Two squares each of side 4 cm are placed side by side to make a rectangle. The rectangle's LENGTH is:

- (A) 4 cm (B) 6 cm
(C) 8 cm (D) 16 cm

Q14. A square park has a perimeter of 40 m. How long is EACH side?

- (A) 4 m (B) 8 m
(C) 10 m (D) 20 m



Q15. Which shape has the GREATER perimeter: a square of side 5 cm, or a rectangle 6 cm by 3 cm?

- (A) The square (B) The rectangle
(C) Both equal (D) Cannot say

Q16. A rectangle has length 9 cm and perimeter 26 cm. What is its BREADTH?

- (A) 4 cm (B) 5 cm
(C) 8 cm (D) 17 cm

Q17. The area of a rectangle is 24 sq cm and its length is 8 cm. Its breadth is:

- (A) 2 cm (B) 3 cm
(C) 4 cm (D) 16 cm

Q18. A wire of length 36 cm is bent into a SQUARE. Each side of the square is:

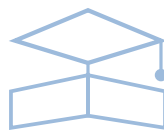
- (A) 6 cm (B) 8 cm
(C) 9 cm (D) 12 cm

Q19. A floor is covered with 20 square tiles, each of area 1 sq m. The area of the floor is:

- (A) 10 sq m (B) 20 sq m
(C) 21 sq m (D) 40 sq m

Q20. Ria walks once all the way around a square garden of side 25 m. How far does she walk?

- (A) 50 m (B) 75 m
(C) 100 m (D) 625 m



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	C	B	C	B	B	C	C	B	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	C	C	C	A	A	B	C	B	C

Step-by-Step Solutions

Q1. Correct answer: **(B)**

Perimeter is the **total length all around the boundary** — like walking once around a park.

Q2. Correct answer: **(C)**

Add all sides: $5 + 6 + 7 = 18$ cm.



Q3. Correct answer: **(B)**

Perimeter = $2 \times (\text{length} + \text{breadth}) = 2 \times (8 + 3) = 2 \times 11 = 22$ cm.

Q4. Correct answer: **(C)**

A square has 4 equal sides: $4 \times 6 = 24$ cm. (36 would be its area.)

Q5. Correct answer: **(B)**

Count the shaded squares: $2 + 2 + 1 = 5$ square units.

Q6. Correct answer: **(B)**

Area is the **space inside** a shape — like the floor space of a room. (The boundary length is the perimeter.)

Q7. Correct answer: **(C)**

Area of a rectangle = length $\times$ breadth = $7 \times 4 = 28$ sq cm.

Q8. Correct answer: **(C)**

Area of a square = side $\times$ side = $5 \times 5 = 25$ sq cm.



Q9. Correct answer: **(B)**

A fence goes AROUND the boundary, so he needs the **perimeter**.

Q10. Correct answer: **(B)**

Covering a surface needs the **area** — the space inside.

Q11. Correct answer: **(C)**

$2 \times (10 + 5) = 2 \times 15 = 30$ m. (50 would be the area: 10×5 .)

Q12. Correct answer: **(C)**

The grid is 3 rows $\times$ 5 columns = **15 square units**. (Multiplying rows $\times$ columns is faster than counting!)

Q13. Correct answer: **(C)**

Two 4-cm squares in a row make a length of $4 + 4 = 8$ cm (and width 4 cm).

Q14. Correct answer: **(C)**

A square's 4 equal sides make the perimeter: side = $40 \div 4 = 10$ m.

**Q15. Correct answer: (A)**

Square: $4 \times 5 = 20$ cm. Rectangle: $2 \times (6+3) = 18$ cm. The **square** has the greater perimeter.

Q16. Correct answer: (A)

Length + breadth = $26 \div 2 = 13$. Breadth = $13 - 9 = 4$ cm.

Q17. Correct answer: (B)

Area = length $\times$ breadth, so breadth = $24 \div 8 = 3$ cm.

Q18. Correct answer: (C)

The wire becomes the perimeter: side = $36 \div 4 = 9$ cm.

Q19. Correct answer: (B)

20 tiles $\times$ 1 sq m each = **20 sq m**.

Q20. Correct answer: (C)

Walking around = perimeter = $4 \times 25 = 100$ m. (625 is the area — a different idea!)

